

Skin Disease Detector – Viva Preparation (CNN, MobileNetV2)

1. CNN (Convolutional Neural Network)

CNN ek deep learning model hota hai jo images ko samajhne aur classify karne ke liye use hota hai. CNN images ke pixels se edges, shapes aur textures automatically seekh leta hai, is liye skin disease detection ke liye best choice hai.

2. Pre-trained CNN & MobileNetV2

Pre-trained model wo hota hai jo pehle hi large dataset par train hota hai. MobileNetV2 lightweight, fast aur kam resources use karta hai, is liye hum ne ise apne project mein use kiya. Hum ne is model ko fine-tune kiya taake skin disease dataset ke mutabiq behtar performance mile.

3. TensorFlow

TensorFlow ek deep learning framework hai jo neural networks ko build, train aur deploy karne ke liye use hota hai. Hum ne TensorFlow aur Keras ka use kiya kyun ke ye CNN models ke liye easy aur efficient hai.

4. .keras Format

.keras TensorFlow ka recommended format hai jo model architecture, weights aur optimizer ko save karta hai. Is format ki wajah se model future mein easily load ho jata hai.

5. Epochs

Epoch ka matlab hai poora dataset ek dafa model se guzarna. Hum ne 20 epochs is liye use ki taake model theek tarah learn kare aur overfitting se bacha rahe.

6. Callbacks

ModelCheckpoint best validation accuracy wala model save karta hai. EarlyStopping overfitting se bachata hai. ReduceLROnPlateau learning rate ko kam karta hai jab validation loss improve nahi hota.

7. Confusion Matrix

Confusion matrix ek table hoti hai jo batati hai kaun si class sahi predict hui aur kaun si ghalat. Rows actual classes aur columns predicted classes hoti hain. Diagonal values correct predictions show karti hain.

8. Supervised Learning

Supervised learning wo technique hai jisme model ko labeled data diya jata hai. Humare project mein har image ka label pehle se known hota hai jaise Acne, Eczema, etc. Loss function supervision ka kaam karta hai aur model apni ghalti sudharta hai.

9. Underfitting

Underfitting tab hoti hai jab model na training data aur na validation data ko theek se samajh pata. Kam epochs ya insufficient training ki wajah se underfitting ho sakti hai.

10. Overfitting

Overfitting tab hoti hai jab model training data ko yaad kar leta hai lekin new data par fail ho jata hai. Hum ne EarlyStopping, validation data aur learning rate control karke overfitting se bachao kiya.

Final Viva Summary

Hum ne skin disease detection system develop kiya jo MobileNetV2 CNN par based hai. Model TensorFlow mein train hua, 20 epochs use huay, callbacks lagaye gaye aur final model .keras format mein save kiya gaya.